SHTPC: Stateful History-to-Payload Closure for Private Heatmap Aggregation

***Abstract*—Private heatmap aggregation over a predeclared client cohort requires authorizing hidden-state transitions without exposing client coordinates to the admission authority. However, composing zero-knowledge (ZK) state proofs with secret-shared aggregation leaves two semantic gaps, as a valid share vector can encode an unauthorized cell and aggregators can release an incomplete or distinguishing subset. This paper introduces Stateful History-to-Payload Closure (SHTPC) to bind hidden-state authorization to retained aggregator shares and complete-cohort first exposure. Exploiting the linearity of one-hot index moments over secret shares, each aggregator signs a masked-moment receipt over its retained share so admission verifies cell equality without in-circuit pseudorandom generator (PRG) seed expansion. Atomic admission and durable reservation further lock the complete eligible cohort before any honest aggregate share becomes decryptable, preserving accepted-history integrity across windows and crash recovery. Against an active aggregator colluding with the Collector, the joint protocol view is simulatable from public metadata and the declared complete-cohort histogram alone. At 10,001 bins, SHTPC requires 5,779 binding constraints versus 79,002,390 for in-circuit seed expansion, accelerating transition proving by 89.8–89.9× over Poseidon vector binding on server hardware and completing in 11.2 s within 0.28 GiB on a Raspberry Pi 4B with under 0.8% binary wire overhead. In 100-client Janus runs, SHTPC incurs 1.44× batch latency (1.38× per session) over plaintext trusted generation while rejecting all tested cell mismatches and subset exposures.**

1. Introduction

A managed fleet with predeclared participants and coordinated sampling uses cohort heatmaps to detect congestion and dispatch vehicles without revealing individual driver trajectories [1–3]. Even when aggregation arithmetic is exact, semantic and membership mismatches distort operational decisions. Redirecting reports away from their history-authorized cells fabricates hotspots, inflating a T-Drive modal cell count from 3 to 22–23 when 20 of 100 reports are substituted in offline replay (Section 5.2), whereas omitting accepted reports suppresses true demand (Figure 1).

Private aggregation validates individual inputs and combines them without revealing any single input. The target application combines an admission authority, which authorizes transitions without reading client coordinates, with two aggregation roles that execute a native secret-shared protocol such as the Distributed Aggregation Protocol (DAP) [4] with

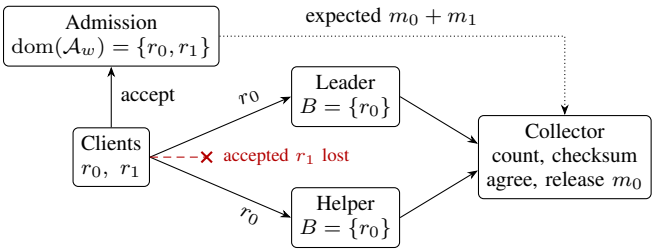


Figure 1. Independent composition can omit a report already accepted by application admission on both role channels. Here $\mathrm{dom}(\mathcal{A}_w)$ denotes the set of report identifiers (IDs) accepted by admission for window w, and B is each role's processed native batch.

Verifiable Distributed Aggregation Functions (VDAFs) [5, 6]. A Collector consumes the resulting cohort statistics. The remaining task is to connect the authorized transitions, retained contributions and release membership to the same accepted history.

Existing protocols validate private inputs [7–10], enforce inclusion or verify aggregates [11–13], and maintain private aggregate state [14]. Each existing mechanism checks its own object in isolation. When an application state proof authorizes cell p while a secret-shared aggregation report encodes a conflicting cell $q \neq p$, independent verification accepts the cross-spliced pair. Superficial identifiers, signed bodies, or hiding commitments associate inputs without binding the retained share vector. Because independent checks accept the Cartesian product of locally valid objects, this vulnerability is structural (formalized in Appendix A).

State freshness and release membership create two further interfaces. A proof for predecessor c remains valid after a competing successor commits, so proof verification does not authorize the current transition. Separately, an accepted report dropped across both receiver channels yields a valid but incomplete batch (Figure 1).

Completeness also has a privacy deadline. Two legal histories can swap two owners' positive cells while retaining the same public metadata and complete heatmap. A subset containing only one of those owners has different aggregates. Once the first honest share becomes decryptable, a corrupted aggregation role and the Collector distinguish the histories, an exposure that later publication refusal cannot retract (Proposition 1). Accepted-history closure connects each admitted claim to its retained contribution and legal state transition, and reconciles the eligible set before first exposure.

Stateful History-to-Payload Closure (SHTPC) binds each

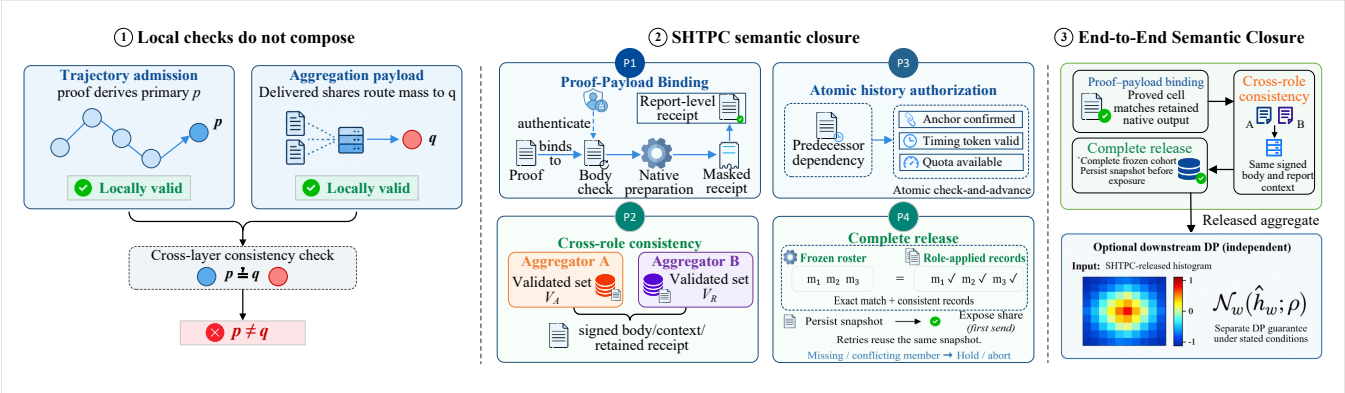


Figure 2. SHTPC connects report meaning, accepted history and complete release through four semantic boundaries.

hidden-state proof to the share vector retained by the Aggregators without reproducing native seed expansion inside the application circuit. Native histogram validity restricts the reconstructed contribution to a one-hot vector with a sentinel. Within this domain, its linear index moment identifies the selected cell. Each role computes a masked receipt from its retained output, and admission compares the paired receipts with the proof-bound masked index. The application circuit is therefore independent of histogram width D, while native vector processing remains $O(D)$.

The bridge introduces receipts and retry feedback that require joint analysis. Authenticated source registration binds private processing to its registered body, and joint simulation accounts for the resulting transcript. Atomic admission advances the current history, while durable reservation fixes the complete cohort before exposure. Figure 2 illustrates the four semantic boundaries connecting client claims to verified release. The threat model separates integrity from privacy, where integrity defends against malicious clients and untrusted transport, whereas privacy tolerates an actively corrupted aggregator colluding with the Collector. Both single-release and predeclared multi-window analyses establish simulation security given equal declared histograms for each permitted complete cohort.

The contributions can be summarized as follows.

1) *Masked-moment binding.* Native one-hot validity and masked receipts from retained outputs bind a hidden-state proof to the actual contribution without reproducing native seed expansion in the application circuit. The standalone relation has 5,779 constraints at 10,001 bins, versus 79,002,390 for direct seed binding, enabling 11.2 s transition proving within 0.28 GiB on a Raspberry Pi 4B with under 0.8% binary wire overhead.
2) *Accepted-history closure.* A security contract connects each admitted claim, retained contribution and legal state transition to complete eligible cohort membership at first exposure. A separation example shows why publication refusal alone is too late. The construction maintains these connections across retries and recovery under honest service execution.
3) *Joint privacy of receipt feedback.* Masked receipts, rejection feedback and retries are jointly simulatable with the native transcript against an active aggregator colluding with the Collector. The simulator receives no aggregate before the first honest share becomes decryptable and one permitted complete-cohort output at that boundary. Simulation distinguishes malicious sources' declarations from their actual retained outputs. Native controls exercise these interfaces, and component and lifecycle measurements report their costs.

The remainder of this paper is organized across six sections. Section 2 formalizes models and contracts. Section 3 details the SHTPC design. Section 4 provides security proofs. Section 5 evaluates integrity and cost. Section 6 surveys related work. Section 7 discusses scope and concludes.

2. Model and Security Contract

2.1. Deployment and Trust Model

Enrolled clients report under stable pseudonymous identities. An admission authority verifies hidden state proofs and maintains accepted history without receiving coordinate openings or role decryption keys. Two Aggregators act as the DAP Leader and Helper [4], denoted $S \in \{A, R\}$ in formulas and as corrupted c and honest h in the privacy analysis. They prepare input shares and retain aggregation outputs. The Collector holds the aggregate-share Hybrid Public Key Encryption (HPKE) key, and its Decoder verifies release records and reconstructs the heatmap. The authority-side Finalizer authenticates the frozen reference and its completion. Its certificate joins that reference to both roles' matching attestations. A durable admission proxy belongs to the same trust domain and delivers signed decisions. Each role durably reserves its own share before export.

Integrity and privacy operate under distinct corruption models. For integrity, clients and network transport are malicious, while admission, both Aggregators, Finalizer, Collector and Decoder follow the protocol using serialized durable storage. This fits a managed fleet enforcing admission and release rules without granting the authority plaintext coordinates. For privacy, one Aggregator may be

actively corrupted in collusion with the Collector, and the adversary may additionally observe the honest authority's business records, receipts, diagnostics and backups (excluding its signing key). Because an active Aggregator can forge shares or abort, this corruption model targets privacy rather than output integrity, with authority corruption remaining out of scope. Forged receipts provide no probing oracle, as admission acceptance for an honest report depends solely on the corrupt role's fixed original share and mask (Section 4.2).

Privacy challenges evaluate pairs of legal histories sharing common public metadata μ, including client identities, task parameters, window indices, step and anchor references, eligibility, contribution bits and delivery schedules. The authority maintains no aggregate query interface, preventing differential reconstruction across intermediate updates. Definition 6 formalizes the single-release privacy contract with one visible cohort per task. Theorem B.3 in Appendix B extends this privacy guarantee to predeclared multi-window executions sharing state across windows. Reports and retained outputs each belong to one window. Later windows can reference earlier accepted states as predecessors or anchors. Privacy bounds protocol leakage beyond the permitted histograms and μ. Activity, contribution bits and cadence remain public, and exact outputs can support inference from auxiliary knowledge.

All guarantees assume correctly sampled public parameters with unknown trapdoors, reviewed fixed-width arithmetic, authenticated confidential registration and durable execution boundaries. Enrollment authenticates pseudonymous identities rather than physical Global Positioning System (GPS) locations or Sybil resistance, continuity is enforced over authenticated slots rather than unverified elapsed time, and authority equivocation lies outside this scope.

2.2. Reports, Histories and Cohorts

Reports move through three distinct units of progress. A sampling slot assigns a measurement to time, an accepted step counts a committed transition, and a release window groups eligible reports. Rejected attempts do not advance the accepted history.

Each report is projected and quantized under fixed public parameters, yielding a hidden vector $\mathbf{x}_r$. A $W \times H$ grid with cell width s selects

$$p_r = \mathsf{Grid}(\mathbf{x}_r) = W \left\lfloor \frac{y_r}{s} \right\rfloor + \left\lfloor \frac{x_r}{s} \right\rfloor, \tag{1}$$

where $D = WH$. The evaluated profile uses $s = 100$ and $W = H = 100$. The replay projection, rounding and boundary rules appear in Appendix A.

All record maps are keyed by report identifier, tracking progress across the three lifecycle stages summarized in Table 1. First, during *registration and locking*, clients register candidate report cores before receiving preparation feedback, establishing the complete registered history map P_w and its eligible projection R_w. Warmup reports, which initialize history before aggregation eligibility, enter log P_w to advance state counters but are excluded from eligible map R_w, whereas eligible zero-bit records ($b_r = 0$) remain in R_w despite adding no mass. Let $I_w = \mathrm{dom}(R_w)$ denote the eligible identifier domain. Honest report identifiers form I_{hon}, partitioning I_w into $I_w^{\mathrm{hon}} = I_w \cap I_{\mathrm{hon}}$ and $I_w^{\mathrm{adv}} = I_w \setminus I_{\mathrm{hon}}$. Because committed records accumulate downstream receipts and commit signatures, projection π_{reg} preserves core registered fields while stripping subsequent receipts and decisions. Map equality compares identical keys and projected values. The concurrent execution log is denoted by $\mathcal{L}$, while H represents an idealized legal sequential history against which concurrent executions are reconciled. Second, during *admission commitment*, reports that pass zero-knowledge (ZK) transition proofs, masked receipts, and atomic history checks are committed, yielding committed log L_w and committed eligible map $\mathcal{A}_w$. Third, during *role application*, roles $S \in \{A, R\}$ process committed decisions into applied maps $L_{S,w}$ and $M_{S,w}$. For each applied report r, role S retains share vector $v_{S,r} \in \mathbb{F}_q^{D+1}$ with a sentinel bin. Each role reserves the sum $\mathbf{a}_{S,w}$ of its retained shares, and decoding the two reserved sums yields the reconstructed D-cell heatmap $\hat{\mathbf{h}}_w$.

TABLE 1. RECORD MAPS AT SUCCESSIVE EXECUTION STAGES.

Stage	All history	Eligible
Registered and locked	P_w	R_w
Admission committed	L_w	$\mathcal{A}_w$
Role S applied	$L_{S,w}$	$M_{S,w}$

For report r, the common admitted object is

$$A_r = (\mathsf{rid}_r, \eta_r, \mathsf{stmt}_r, \mathsf{pay}_r, c_{r-1}, c_r, w, b_r). \tag{2}$$

The context η_r binds identity, window and policy. The remaining fields identify the proof statement, bound channel packages, predecessor, successor and contribution bit. A complete signed body, two retained-output receipts and one commit decision authenticate this object in the native realization (Appendix B).

Strict release produces the complete eligible cohort or refuses, serving applications that require coordinated sampling and can tolerate delayed batch release. Execution separates failures across three distinct phases. (1) *Pre-lock registration failure* occurs when a client fails to submit a complete signed body or assign a required contiguous slot before the registration lock boundary. (2) *Post-registration validation stall* occurs when an installed body contains an invalid proof, references an uncommitted anchor, or violates quota. (3) *Post-commit transport failure* occurs when network disconnection happens after an admission decision is durably committed. Once a complete request is registered and committed, subsequent client disconnection does not halt cohort processing. Server-side durable redelivery and idempotent retry continue role application and export autonomously without continuous client presence. A member permanently missing during Phase (1) registration or Phase (2) validation withholds release, prohibiting dynamic member dropouts

TABLE 2. NATIVE WINDOW LIFECYCLE AND SCOPE.

Stage	Contract
Register	All predeclared user identifiers (UIDs) and contiguous step slots, including warmups and eligible zeros, require complete pending bodies.
Lock	At the registration boundary, the complete graph starts at actual heads. Both roles install the lock.
Process	Current-head permits precede preparation and commit. Both roles apply decisions.
Expose	Complete records are reserved before the first share. Retries use the same snapshot.
Abort	Partial commits and quota persist. Later configured slots and new bodies start at actual heads.

or publishing over the receiver intersection set. Table 2 summarizes the window lifecycle.

2.3. Accepted-History Closure

Four obligations share A_r. Payload binding (P1) connects the proved claim to the consumed package by

$$\mathsf{Contrib}(\mathsf{pay}_{r,\mathrm{consumed}}) = b_r \mathbf{e}_{p_r}. \tag{3}$$

Channel consistency (P2) requires both roles to consume channels of that same object. Atomic transition (P3) gives each accepted object one atomic transition with its quota charge,

$$\mathrm{State}_{r-1} \to \mathrm{State}_r. \tag{4}$$

Complete release (P4) requires a successful release to contain exactly the eligible committed records and the complete locked reference,

$$\pi_{\mathrm{reg}}(\mathcal{A}_w) = R_w. \tag{5}$$

These obligations organize the successful-trace and exposure-prefix definitions below. Theorem 1 establishes their composition under the stated integrity premises.

Definition 1 (Legal sequential history)**.** A sequence $H = (A_1, \ldots, A_n)$ with unique identifiers is legal if replay from enrolled genesis enables every transition. Each step has an authorized owner and context, names the current head and policy-selected accepted anchor, and satisfies the application relation on opened states. It atomically appends the successor, advances the head and accepted-step counter, and charges quota without changing other owners. Rejected and uncommitted attempts add no step.

Definition 2 (History refinement)**.** For a finite prefix, $\mathcal{L} \preceq H$ when committed admission operations bijectively map to a legal H, preserving every A_r field and the order of linearization points between invocation and response. Committed pending operations are completed at the prefix end, while uncommitted attempts are discarded. Retries of one durable decision represent one operation, and distinct committed effects remain distinct.

Definition 3 (Successful-trace closure)**.** Fix a transition policy, additive contribution map $r \mapsto m_r$ and eligibility rule.

TABLE 3. MISSING INTERFACES, COUNTEREXAMPLES AND SUPPORTING CHECKS.

Correct mechanisms	Open interface	Counterexample and check
Proof and native input validity	Proved value equals the retained output	Cross-splicing counterexample (App. A). Native mismatch rejected (Sec. 5.3)
Report identifier and signatures	Authenticated source before private processing	Ciphertext-transplant attack. Native guard-off leak and guard-on rejection (Sec. 5.4)
Proof verification	Head and quota recheck at commit	Stale-head or duplicate-retry counterexample. Native trace and bounded model (Sec. 5.5)
Admission and set checking	Complete set fixed before the first decryptable share	Subset separation (Prop. 1). Native seven-of-eight exposure blocked (Sec. 5.4)
Set checking	Durable reservation before send	Post-send crash counterexample. Persistence premise (Sec. 3.6)

A successful release has closure if some legal H satisfies $\mathcal{L} \preceq H$. Let $E_w(H)$ be the set of records in the complete accepted history H that belong to window w and satisfy the eligibility rule. Their identifiers equal $\mathrm{dom}(\mathcal{A}_w)$, and each identifies one common admitted object across proof, packages, transition and certificate. Each role's application count $N_{S,w}(r)$ satisfies $N_{S,w}(r) = \mathbf{1}[r \in \mathrm{dom}(\mathcal{A}_w)]$, and its reserved sum uses exactly those retained outputs. The pre-noise output is $\sum_{r \in \mathrm{dom}(\mathcal{A}_w)} m_r$. For heatmaps, $m_r = b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}$. Eligibility is independent of bit value.

Definition 4 (Exposure-prefix safety)**.** Every decryptable honest share comes from a prior durable reservation satisfying

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w = \pi_{\mathrm{reg}}(\mathcal{A}_w),$$
$$\mathbf{a}_{S,w} = \sum_{r \in I_w} v_{S,r}. \tag{6}$$

Retries preserve its membership and plaintext. Successful-trace closure also requires correct final reconstruction.

Table 3 identifies missing interfaces and separates their counterexamples from native controls, bounded models and execution premises. Appendix A formalizes cross-splicing and identical-receiver-view arguments.

3. SHTPC Design

SHTPC realizes P1–P4 by connecting hidden-state authorization to retained aggregation outputs and complete first exposure.

3.1. Design Overview

Anchor Distance Window Continuity (ADWC) bounds each hidden move relative to the last accepted position and a policy-selected historical anchor. The client proves ADWC and grid membership over hidden states. Native validity and masked-moment receipts connect the proved cell to the outputs retained by the aggregation roles. Admission checks these receipts and authorizes their use against the current accepted history. The workflow below connects admission to complete release.

3.2. End-to-End Workflow

Enrollment fixes owner keys, a proved hidden genesis and the task configuration. Each window then proceeds through five steps.

1) *Lock.* Clients register complete signed bodies with admission. The authority verifies the candidate graph and locks P_w and R_w before private preparation or receipt feedback.
2) *Retain.* Both roles install the lock. A permit authorizes preparation against the current head and anchor. Each role prepares its input, retains its native output and signs a masked-moment receipt.
3) *Commit.* Admission checks both receipts and their masked-moment equality. A transaction rechecks the actual head, anchor and quota, then stores the successor, history and signed decision together.
4) *Apply.* A durable proxy delivers the decision to each role. Each role applies it once and updates its retained history and eligible map. Warmups advance history without entering the eligible map.
5) *Reserve.* Each role checks its applied records against the frozen reference and verifies the retained-output links. It durably reserves the complete sum and selector before sending its share. Retries use the same snapshot.

3.3. Binding Hidden State to Native Outputs

P1 binds the state-authorized cell to the retained native output without encoding native pseudorandom generator (PRG) seed expansion inside the ZK circuit. Under native private aggregation, the underlying verifiable aggregation layer enforces one-hot validity of the reconstructed vector ([5, Section 7.4.4]). For roles A and R retaining shares $v_A, v_R \in \mathbb{F}_q^{D+1}$, the reconstructed output satisfies $v = v_A + v_R = \mathbf{e}_y$ for a unique unit coordinate $y \in [0, D]$.

Over this one-hot domain, the linear index moment $\sum_{j=0}^{D} jv[j] = y$ uniquely identifies the selected index y (for example, in a three-cell vector $(0,0,1)$, $0{\cdot}0+1{\cdot}0+2{\cdot}1 = 2$). In native field $\mathbb{F}_q$ with $D < q$, let a state-derived primary cell $p < D$ and public bit b authorize target index $x = bp + (1-b)D$, with sentinel D. The client samples private uniform masks $r_A, r_R \in \mathbb{F}_q$, commits to their openings, and encrypts r_S for role $S \in \{A, R\}$. The ZK proof binds x and the masks to the public target

$$T = x + r_A + r_R \pmod q. \tag{7}$$

TABLE 4. BINDING CHOICES AND THEIR CONNECTION TO THE NATIVE CONTRIBUTION.

Approach	Certified connection	Additional work or condition
Association only	Common report or committed value	Retained-output equality still needed
Direct seed binding	Seed expansion and selected vector	Native sharding in the circuit
Generic vector binding	Retained shares equal certified vector	Retained-share vector commitment
SHTPC (this work)	Certified index equals retained index	Native one-hot validity

Each correct role computes its masked moment from the retained output as

$$m_S = \sum_{j=0}^{D} jv_S[j] + r_S \pmod q. \tag{8}$$

By linearity, summing the two receipts gives

$$\begin{aligned} m_A + m_R &= \sum_{j=0}^{D} j(v_A[j] + v_R[j]) + r_A + r_R \\ &= y + r_A + r_R \pmod q. \end{aligned} \tag{9}$$

Consequently, checking $m_A + m_R = T$ directly tests $y = x \pmod q$. Lemma 1 shows that acceptance forces the reconstructed contribution to equal $b\mathbf{e}_p + (1-b)\mathbf{e}_D$, recovering $b\mathbf{e}_p$ upon sentinel removal. Table 4 compares this approach against generic vector binding and direct seed binding.

Transition circuit relation (ADWC and grid). The public statement binds the state commitments, report context and masked index. Its witness opens the state, ticket, mask and receipt commitments. The relation binds the payload witness's cell and remainder to the successor and enforces $p = \mathsf{Grid}(\mathbf{x}_r)$, $\|\mathbf{x}_r - \mathbf{x}_{\mathrm{prev}}\|^2 \le \tau_u^2 \Delta t_t^2$ and $\|\mathbf{x}_r - \mathbf{a}_t\|^2 \le \min\{\tau_u^2 (\Delta T_t^{(K)})^2, c_{\mathrm{policy}}^2\}$. Here $\mathbf{a}_t$ is the policy-selected accepted anchor at lag K, Δt_t and $\Delta T_t^{(K)}$ are the slot-time gaps to the predecessor and the anchor, τ_u is the speed tier and c_{policy} is the anchor cap. Admission checks that the referenced predecessor and anchor are the actual accepted ones. The same grid and distance constraints apply when $b = 0$. Appendix A gives the parameters, and the artifact's fixed transition circuit gives the ordered public-signal interface (Section 8).

Cross-field canonical encoding ($\mathbb{F}_q$ vs. p_{proof}). To bridge native field $\mathbb{F}_q$ and proof scalar field p_{proof}, canonical scalars enforce $x + r_A + r_R = T + \chi q$ with carry $\chi \in \{0,1,2\}$ over integers, preventing wrap-around between disparate field moduli (Appendix B). State commitments retain birth domains and references, and admission verifies the masked-moment sum without receiving coordinate openings or Aggregator decryption keys.

Complexity and idempotent preparation. Because moment checking operates on scalars while share evaluation scans $D+1$ buckets, the application circuit size is independent of histogram width D, whereas native vector pro-

cessing remains $O(D)$. Separating share preparation from accumulation ensures that each signed receipt authenticates the package context and retained share, with exact retries replaying the same output and receipt.

The compact binding relation relies on the injectivity of index moments over one-hot vectors $\mathbf{e}_y$. While a first-order moment is non-injective on general vectors (for example, $(0, 2, 0)$ and $(1, 0, 1)$ both evaluate to 2), multi-dimensional or weighted stateful applications can apply higher-order or component-wise moments whenever their reconstructed domain admits an injective moment relation.

3.4. Protecting the Feedback Interface

Encryption does not authenticate an input's owner. An attacker can transplant an honest ciphertext and substitute a known mask, causing an honest receipt to reveal the cell before admission rejects the report. This feedback arises from the new bridge even when the proof and native protocol are secure in isolation.

SHTPC allocates an owner identifier to one complete signed body and registers it before private processing. Each ingress authenticates the consumed metadata, ciphertexts, sidecars, installed lock and permit against that body. Both retained sessions and receipts therefore refer to the same admitted object, realizing P2. Source registration, origin locking, and fixed original shares prevent an active adversary from splicing components across reports or adapting receipts to probe honest state. Because the registered body fixes the original share and mask, and retry and abort states are serialized, adaptive feedback cannot alter the receipt acceptance condition.

Recovery retains the same source and native state. The Helper journals its first protected input and response before sending. Identical retries replay those bytes, while changed inputs or new jobs cannot reopen that branch. The Leader's lease and transaction fence permit at most one committed transition. Uncommitted computation can receive new peer feedback, which the joint privacy analysis includes explicitly.

3.5. Atomic History Authorization

A transition proof can remain valid after another successor commits. Admission therefore verifies the proof before the transaction, then atomically rechecks the actual head, anchor and quota and stores the successor, history and signed decision. Both retained-output checks precede this linearization point. Preparation, admission and role application use separate databases connected by that decision and its output links.

For stable identity u, task and window w, durable quota $q_{u,w} \in \{0, 1\}$ is shared across credential epochs. With accepted step t and $N_{\mathrm{warm}} = \max(K, 2)$,

$$\begin{aligned} \gamma_t &= b_t \mathbf{1}[t > N_{\mathrm{warm}}], \\ q_{u,w} + \gamma_t \le 1, &\qquad q_{u,w} \leftarrow q_{u,w} + \gamma_t. \end{aligned} \tag{10}$$

Commit induction gives

$$q_{u,w} = \sum_{\substack{r \in \mathcal{A}_w \\ \mathsf{uid}(r) = u}} b_r \le 1. \tag{11}$$

Warmups remain in the transition log without charge. Eligible zero-bit updates remain members and can advance state after quota is spent.

History is keyed by stable $(\mathsf{task}, \mathsf{uid})$, with credential, revision, window, receipt and budget checks inside the transaction. Credential epochs and storage revisions do not replace accepted-step counters or sampling slots. The proxy queues exact decisions durably before delivery. Role databases apply each decision once, including out-of-order delivery. Retries return that decision, and credential rotation preserves history and spent quota.

3.6. Complete Exposure and Recovery

P4 fixes membership before preparation feedback can influence it. Admission locks the verified candidate graph and its eligible projection at the public registration boundary. The graph includes warmups and eligible zeros together with their predecessor and anchor references. Preparation errors and receipts cannot select a smaller cohort or replace a locked body. Export by role S requires

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w. \tag{12}$$

These equalities compare the frozen registration cores, including context and body. The role separately checks each actual retained-output link against its authenticated receipt and commit decision.

In one export transaction, the role verifies Eq. (12) and reserves the selector, record map, retained-output links, and exact cohort sum (under the strict profile) before sending its share. Because role application is monotone ($0 \to 1$), this atomic check refines Definition 3 by ensuring all locked members are applied before exposure (Appendix A). Persisting the resulting snapshot before transmission ensures that re-exports idempotently re-emit the identical complete sum without altering membership. Warmup completion is checked through $L_{S,w}$, and eligible zeros remain in $M_{S,w}$. Numerical equality alone cannot establish this membership because an omitted eligible zero has no displayed mass. Responses and re-encrypted retries read the same durable snapshot, so recovery cannot change its plaintext or membership. The Finalizer authenticates the frozen reference and both roles' matching attestations. The Decoder checks that certificate and the reserved records before reconstruction.

After locking, any member failing registration or validation holds or aborts the cohort without silent exclusion or intersection release (Section 2.2). Conversely, committed decisions survive client disconnection via autonomous server-side redelivery without re-upload. Collection failure preserves the first exposure and reserved snapshot. Pre-exposure abort retains partial commits, heads and quota for continuation as in Table 2. For example, owners left at steps 2 and 1 resume at 3 and 2 with fresh bodies referencing

TABLE 5. SECURITY PROPERTIES AND OBSERVER VIEWS RELY ON DISTINCT MECHANISMS AND PREMISES.

Property or observer	Main mechanism	Premise or permitted output
Legal hidden witness	Groth16 [15] extraction, Pedersen [16] binding	Unknown trapdoors and typed generic-group extraction
Proved cell is counted	One-hot validity, masked moments	Correct retained outputs and $D < q$
Legal history step	Authentication, commit induction	Atomic head and quota update with durability
Complete first exposure	Lock, apply and reserve	Lock before feedback and persist before send
Authority privacy	Blinding, ZK, HPKE and receipt simulation	Common metadata and no aggregate interface
Role and Collector privacy	Full-verifier and joint-view simulation	Static corruption, one cohort and equal full sums
Combined observer	Joint business-view simulation	Same complete output and no authority signing key

those committed heads. Incompatible client schedules are rejected at the lock boundary, ensuring that task configuration remains durably bound to the database. Theorem B.3 in Appendix B establishes the privacy conditions for predeclared windows.

For a finite fully committed lock, eventual service and channel recovery with fair retries and no deadline stop gives complete release. Each successful apply removes one of at most $2|P_w|$ pending role-record pairs. Permanent client refusal and incomplete pre-lock registration fall outside these progress conditions. Admission caps bound accepted work and exact retries consume no new slot. The native profile uses $B_u = 16$ per owner and window.

4. Security Analysis

The integrity argument connects three steps. Extraction identifies the certified witness, and the moment check binds it to the retained contribution. Atomic admission with durable reservation then extends this binding to first exposure. The privacy argument separately simulates the receipts and native transcript, using the permitted complete-cohort histogram at first exposure. Full symbolic security ledgers, concrete hybrid sequences and reduction proofs are provided in the anonymous artifact repository (indexed in Section 8).

4.1. Integrity and First Exposure

Table 5 separates the premises for history closure from those for observer privacy.

Definition 5 (Closure game)**.** The challenger initializes honest service keys, enrolled identities and genesis state. A bounded probabilistic polynomial-time (PPT) adversary controls malicious clients and delivery. It submits packages, observes responses, replays, replaces, reorders or suppresses messages, and invokes admission and window closure. Honest signing keys and committed service state remain outside its control. The challenger records the locked P_w, R_w, invocation and response events, committed objects and receiver application counts. The adversary wins if the Decoder accepts a release for which no legal linearization exists, or any membership, application-count, common-object or reconstruction condition of Definition 3 fails, or if a native role exports a share that violates Eq. (6). Refusal without an unsafe export is not a win. The game fixes the strict profile. Write $\mathrm{Adv}_{\mathrm{closure}}(\mathcal{B}) = \Pr[\mathsf{Win}]$. Serializable durable state and honest service execution are premises of this game.

Lemma 1 (Native moment fidelity)**.** *Assume sound relation certification, commitment binding, authenticated common body identity, correct receivers and native validity. Let the certified canonical coordinate be $\mathbf{x}_r$, with $p = \mathsf{Grid}(\mathbf{x}_r) < D < q$ and $b \in \{0,1\}$. Let $x = bp + (1-b)D$, with canonical $r_A, r_R, T \in [0, q-1]$, and $F(v) = \sum_{j=0}^{D} jv[j]$. Suppose the retained outputs satisfy $v_A + v_R = \mathbf{e}_y$ and the receipts satisfy $m_S = F(v_S) + r_S$. If certification fixes $T = x + r_A + r_R \pmod q$ and admission checks $m_A + m_R = T$, the reconstructed contribution after sentinel removal is $b\mathbf{e}_p$.*

Proof. Opening checks fix the certified masks. Linearity gives $y = F(v_A + v_R) = T - r_A - r_R = x \pmod q$. Since $x, y \in [0, D]$ and $D < q$, this is integer equality, and removing sentinel D yields $b\mathbf{e}_p$. □

Proposition 1 (First-exposure separation)**.** *Under the share-observation interface of Section 2.1, suppose two metadata-compatible admissible histories have the same aggregate on a full accepted set $\mathcal{C}$, but different aggregates on a permitted native batch $R \subsetneq \mathcal{C}$. Native validity, atomic admission and exact-set checking at final publication do not imply exposure-prefix safety or privacy relative to the full aggregate. This holds even for a wrapper that successfully publishes $\mathcal{C}$ on complete delivery.*

The separating execution delivers complementary shares of R to the corrupted role and Collector, then refuses publication because $R \neq \mathcal{C}$. As constructed in Appendix A, this exposure distinguishes the legal histories despite subsequent publication refusal.

Proof interface. The privacy simulation generates proofs for honest sources, while an adversary can reuse real or simulated proofs and combine material from related setup keys. A single-proof knowledge-soundness statement does not specify this source-reuse interface. Lemma 2 supplies the required alternative. An accepted proof yields a witness for a new statement or refers to a registered source with the same semantics. Lemma 1 then binds the contribution, and atomic history and durable export give the event induction in Appendix A.

Related-setup extraction. The analysis uses typed generic bilinear groups without a target-to-source map, correct se-

tups with unknown trapdoors, independent latent variables and ceremonies with at least one honest contribution each, nonzero independent phase-two scalars, and finite source and oracle budgets. Correlated key points retain their setup relations. Only the current genesis and transition namespaces are accepted. Auxiliary keys remain visible to the adversary. The inventory and allowed source encodings are specified in Appendix B.

Statements are registered before fresh proof coins by key namespace, relation and ordered public signals. Honest-source reuse keeps its registered meaning. Known or extracted witnesses are cached, and malicious owners receive no witness-free proof service.

Lemma 2 (Related-setup extraction)**.** *Under these source rules, every accepted proof yields a witness or names a registered source with identical key, relation and ordered public signals, except with probability at most g. Appendix B gives its finite budget.*

Proof intuition. The joint source representation tracks real and simulated proofs under each key. Comparing the target verification equation separates registered target-source reuse from fresh statements. For fresh statements, generic bilinear group divisibility forces the proof elements to encode a valid assignment for the target relation, extracting the witness. Unequal formal expressions are charged to the generic group collision bound. □

Atomic commits witness Definition 2. Durable receiver indexes enforce the application counts in Definition 3. Appendix A connects both invariants to export.

Theorem 1 (Accepted-history composition at first exposure)**.** *In Definition 5 under Section 2.1, let B_C bound the combined probability of whole-execution proof extraction, commitment binding, channel-package authentication, digest collision, and origin guessing failures, and let v_N bound native validity failure. Correct serialized durable execution is a premise. For the verification-key tag in v_N, the integrity instantiation adopts the one-wayness assumption for Secure Hash Algorithm 256 (SHA256) on uniform 128-bit keys with resource-bounded advantage a_{OW128}. Then*

$$\mathrm{Adv}_{\mathrm{closure}} \le \min\{1, B_C + v_N\}. \tag{13}$$

Concrete parameterizations, query budgets, and algebraic expansions for B_C and v_N appear in Appendix B. Outside those failure events, every first export and retry satisfies Eq. (6) *with one immutable snapshot. The accepted log refines a legal ADWC history under Definitions 1 and 2. Every successful strict release satisfies Definition 3 and reconstructs*

$$\hat{\mathbf{h}}_w = \sum_{r \in I_w} b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}. \tag{14}$$

Each record has one common admission object across its proof, consumed packages, committed transition, and release certificate. These invariants persist through recovery and collection failure.

Proof sketch. Under the source rules of Lemma 2, honest same-source reuse retains its registered semantics, malicious sources use known or extracted witnesses, and foreign namespaces or contexts are rejected. Commitment binding fixes the predecessor and anchor. Native validity and Lemma 1 bind its retained output to the proved cell. Induction over registration, lock, permit, commit and apply preserves the fixed candidate maps and a legal accepted history. The export transaction checks full completion and fixes the share before sending. Recovery preserves that transaction's snapshot. The roles apply the durable certificate idempotently without a cross-database atomic commit. After a correct freeze, complete membership is stable under monotone application, so an earlier incomplete snapshot can refuse but cannot authorize a smaller set. Consequently every exposure prefix satisfies Eq. (6), and reconstruction gives Eq. (14). Appendix A gives the event cases. □

Equation (14) also supplies the admitted-history numeric input to any downstream mechanism, without changing that mechanism's own privacy or output policy.

4.2. Joint Privacy with Feedback

Definition 6 (Single-cohort privacy experiment)**.** Fix a finite public resource budget before setup. Sample a correct setup and fix the task, profile, keys and static corruption set before any honest measurement. The budget covers honest and adversarial work, including warmups, zero-bit reports, failed and unprepared reports, verification, oracle calls, bytes and retries. Execution stops at the budget. The environment supplies two complete legal histories H_0, H_1 for honest sources with the same public skeleton μ. It fixes each source's challenge inputs before its first report, while the adversary can adaptively request reports from those histories. Honest reports reach admission over an independent authenticated confidential channel according to the input-ready plan. Local witness and proof generation time is outside the online view.

For $X \in \{A, P, P{+}A\}$, the challenger selects a uniform hidden bit d, runs the protocol on H_d, and interacts with a bounded PPT adversary $\mathcal{B}$ with the following view. In A, the correct authority's view includes its business records, receipts, diagnostics, database and backups, and its ordinary authorization secrets. It has no role decryption keys or aggregate output. The observer may compute offline with its keys but cannot use them to issue extra authority signatures, change the configuration or bypass the state machine. In P, one actively corrupted Aggregator and the Collector share their state, decrypted inputs, peer messages and actual available Collector results. The corrupted role can fabricate messages and shares or abort. In $P{+}A$, that observer also sees the correct authority's business view, without its authorization secret or malicious issuance. The adversary can control corrupted owners and schedule, delay, replay or drop messages within the model. Each view includes its visible proofs, ciphertexts, commitments, responses, refusals, retries, aborts and modeled online times. These outcomes are not assumed to be part of μ.

For each record, let $y_i^{(d)} = b_i p_i^{(d)} + (1-b_i)D$ denote its declared native index, where $p_i^{(d)}$ is its declared primary cell. For each allowed complete cohort R_w, the ideal interface returns the declared complete-cohort histogram $\mathcal{O}_d(R_w) = \sum_{i \in I_w} \mathbf{e}_{y_i^{(d)}}$, including sentinel D. Write $\mathcal{O}(R_w)$ when the world is clear. Because a corrupted role or source can alter shares, $\mathcal{O}_d(R_w)$ sums declared indices rather than the Collector's reconstructed output. The A simulator receives no aggregate query and its two histories need not have equal histograms. The P and $P{+}A$ histories have equal declared complete-cohort histograms for every allowed complete cohort. Their simulators may query that full cohort once at the boundary where the first honest share becomes decryptable, even if collection later fails. This exposure-time interface returns no output before that boundary. Retries reuse the reserved plaintext without another query. No subset query is permitted. The game covers one visible selector for the fixed task and key.

The adversary outputs a bit in Exp_X^d. Its distinguishing advantage is $\mathrm{Adv}_X(\mathcal{B}) = |\Pr[\mathsf{Exp}_X^1(\mathcal{B}) = 1] - \Pr[\mathsf{Exp}_X^0(\mathcal{B}) = 1]|$.

Theorem B.3 in Appendix B uses a private-prefetch variant of this interface. After durable reservation of a complete cohort, send preparation can query its output before the first honest share becomes decryptable. The answer remains private if preparation aborts without sending. The proof couples this early evaluation with exposure-time evaluation, preserving observable prefixes, including executions that never expose a share. Recovery retains the same cohort and cached answer, maintaining indistinguishability across aborted runs.

Theorem 2 (Joint privacy of admission and aggregation views)**.** *Fix the privacy experiment of Definition 6 with static corruption, the strict profile, and the source and programmable-oracle models in Appendix B. The authority view is simulatable without honest coordinates or an aggregate. The aggregation view, including any shared authority business view, is simulatable from the observer's inputs, μ and the permitted complete-cohort histogram. The distinguishing advantage of each observer X is at most $\mathrm{Adv}_X \le \min\{1, 2e_X\}$, where e_X composes the primitive bounds above, the hidden HPKE encapsulations and the native stream simulation. Theorem B.2 in Appendix B gives the concrete form and a hybrid outline.*

The masked receipts provide the key link. Let c be the actively corrupted role and h the honest role. For an honest source report, the honest role computes $m_h = F(v_h^{\mathrm{orig}}) + r_h$ over its retained share, where $F(v) = \sum_j j v[j]$. Admission accepts paired receipts (m_h, z) if and only if $m_h + z = T$. Because $T = x + r_h + r_c \pmod q$ and native one-hot validity gives $F(v_h^{\mathrm{orig}}) + F(v_c^{\mathrm{orig}}) = x$, an arbitrary corrupt receipt z passes admission if and only if

$$\begin{aligned} z = T - m_h &= (x + r_h + r_c) - (F(v_h^{\mathrm{orig}}) + r_h) \\ &= F(v_c^{\mathrm{orig}}) + r_c \pmod q. \end{aligned} \tag{15}$$

Whether z passes therefore depends only on $F(v_c^{\mathrm{orig}}) + r_c$, which the corrupt role's own input share and mask determine.

This refusal predicate establishes that malicious feedback cannot serve as a probing oracle under the joint distribution $(v_c^{\mathrm{orig}}, r_c, T, m_h, \mathsf{status})$. This invariance is not a heuristic appeal to the marginal uniformity of r_h. Rather, authenticated source registration binds the complete client package prior to role processing, locking v_c^{orig} and r_c to the immutable body. Idempotent retries and linearized commits evaluate against this same fixed predicate. Because x, v_h^{orig}, and r_h cancel out in Eq. (15), admission outcomes reveal no information about honest trajectories beyond the corrupt role's own inputs.

For the simulator, uniform T determines the coupled honest receipt in Eq. (15) prior to learning the cell. For the authority alone, uniform T, U give $(m_A, m_R) = (U, T-U)$ without an aggregate query. These identities join the receipt simulation to the native preparation transcript and the same causal history state.

The simulator first classifies each request by its registered body and exact statement, including the key namespace and ordered public signals. Honest same-source reuse preserves registered business semantics and its native handle. Malicious sources use known or extracted witnesses. Foreign namespaces or contexts are rejected. A changed honest body under the same statement requires a digest collision or an unauthorized signature. For a fresh statement, extraction supplies the committed openings, not the ciphertext plaintext. A ciphertext distinct from the protected challenge tuple is handled by legitimate decryption and native execution. This routing permits malicious sources without providing an oracle for an honest source's challenge.

For an honest source, the coupled receipt in Eq. (15) uses the same native share as the preparation transcript. For a malicious source, let y_j be its extracted or known claimed index and $w_{h,j}$ the actual successful output of the honest role. The simulator does not assume that these outputs satisfy the honest-source identity. When reservation permits first exposure, the single complete-cohort query $\mathcal{O}(R_w)$ gives

$$\begin{aligned} V_{h,R_w} = \mathcal{O}(R_w) &- \sum_{j \in I_w^{\mathrm{adv}}} \mathbf{e}_{y_j} \\ &- \sum_{i \in I_w^{\mathrm{hon}}} v_{c,i}^{\mathrm{orig}} + \sum_{j \in I_w^{\mathrm{adv}}} w_{h,j}. \end{aligned} \tag{16}$$

Here I_w^{hon} and I_w^{adv} partition the eligible report identifiers. Subtracting the known or extracted malicious-source declarations leaves the honest-source sum. For each honest source, $v_{h,i} = \mathbf{e}_{y_i} - v_{c,i}^{\mathrm{orig}}$. Subtracting the original corrupt-role shares therefore gives their honest-role aggregate over honest reports. Adding the actual honest-role outputs $w_{h,j}$ for malicious sources completes the joint view without subset queries or assuming that those sources follow their declarations. Retries reuse the reserved plaintext. The authority-only simulator uses the zero-query receipt law above.

Appendix B gives the Histogram kernel and reduction budgets for the joint simulation.

5. Evaluation

The evaluation examines trajectory impact, binding, first exposure, recovery and the cost of the enforced contract.

5.1. Setup and Comparison Contract

Datasets and workloads. The evaluation exercises empirical and synthetic mobility workloads. Offline replay aligns 100 selected legal histories per dataset by relative step across three fixed actor permutations, using 300-second sampling slots and a 100-m grid spanning 10 km (Appendix A).

1) *T-Drive [17].* GPS trajectories from Beijing taxis.
2) *Porto [18].* Trip trajectories from Porto taxis.

Implementation and setup.

1) *Protocol.* SHTPC, denoted Full in system runs, uses Janus [19] with two-role Prio3Histogram under the DAP specification [5, 20]. The histogram contains 10,000 spatial cells and a sentinel.
2) *Hardware.* Server benchmarks and Full/T runs use an Intel Xeon Gold 6148 central processing unit (CPU) at 2.40 GHz (Node.js 22.22.1, snarkjs 0.7.4). Edge client benchmarks use a 1.8 GiB Raspberry Pi 4B with a 1.80 GHz ARM Cortex-A72 CPU (Node.js 20.19.2, Rust 1.85.0).
3) *Full/T workload.* Five paired runs evaluate synthetic geometry with fresh identities and keys, covering 100 genesis proofs, six priming steps per owner, and one eligible step per owner. Each measured release contains one nonzero eligible report per owner, while the six preceding steps initialize history rather than contributing to that release. The synthetic batch operates without an artificial sampling delay. Client generation uses 16 workers, queuing up to 64 bodies with at most four bodies per public drive call.

Baselines and controls.

1) *Binding alternatives.* Evaluation compares against direct in-circuit seed expansion and a Poseidon-based share-vector binding circuit [21].
2) *Trusted generation (T).* A plaintext-trusted baseline retains Prio3 report generation, preparation, aggregation, registration, freeze, durability and collection, but replaces ZK state transition proofs and masked-moment checks with direct plaintext report synthesis from coordinates, state and randomness while retaining genesis proofs. Signed configuration selects T receipts or Full proof and sidecar verification, and eight-client controls reject cross-mode or tampered inputs.
3) *SCIF and PEAR.* An acceptance bridge is evaluated for SCIF [11] alongside an eight-report validity run for PEAR [12] under their respective aggregation interfaces.
4) *Offline oracle.* Replay compares independent composition with a strict-contract oracle that detects every inconsistency.

Batch execution. The timed workload processes available inputs as a fixed-cohort batch, registering and locking all candidate bodies before native preparation. The 300-second slots belong to trajectory replay rather than a processing deadline for the synthetic Full/T run.

Metrics.

1) *Replay.* Cohort-normalized L_1 error and target-cell counts measure spatial distortion.
2) *Native controls.* Acceptance, release and recovery report observed outcomes.
3) *Component and system costs.* Binding benchmarks use three matched inputs, one logical CPU core, fresh Node processes, one excluded warmup per relation and input, and five interleaved pairs per input. System endpoints measure the client-generation-through-collection window and full session.

5.2. Real-Trajectory Impact and Native Controls

The evaluation examines three complementary workloads. First, $N = 100$ real-trajectory offline replay quantifies hotspot distortion under report substitution or omission. Second, $N = 8$ live Janus controls on empirical traces verify end-to-end service integration and attack refusal. Third, $N = 100$ (700-transition) synthetic runs measure pipeline cost and recovery (Sections 5.5–5.6). Because transition circuit constraints and Prio3 share processing are invariant to specific coordinate values within the legal grid (Table 6), synthetic legal trajectories exercise the identical cryptographic and database execution paths as empirical traces.

Offline impact. Substituted reports can sharply amplify an apparent hotspot. In offline replay, replacing 20 of 100 reports produces 0.38–0.40 cohort-normalized L_1 error, raising the clean modal cell count from 3 to 22–23 for T-Drive and from 2 to 21–22 for Porto. The strict-contract oracle withholds every tested window with nonzero substitution or omission, trading availability for integrity.

Native Janus controls. Janus runs over the first eight selected pressure histories (Appendix A) from T-Drive and Porto release exact clean outputs and refuse both substituted reports in each 25% attack run before commit, preventing incomplete-set export.

Trajectory impact. Replacing 20 of 100 reports inflates modal hotspot counts from 2–3 to 21–23 (0.38–0.40 normalized L_1 error) in offline replay, while eight-client Janus runs reject all substituted reports before commit.

5.3. Compact Semantic Binding

When a native-valid report encodes cell 42 while its ZK proof authorizes cell 2312, the masked-moment check rejects the report without a commit, head advance or quota consumption, validating Lemma 1.

At 10,001 bins, the standalone masked-moment relation requires 5,779 constraints, compared with 79,002,390 for in-circuit seed expansion, representing a 13,670.60× reduction (Figure 3). The relation leverages native one-hot validity,

TABLE 6. SAME-INPUT TRANSITION COSTS AT 10,001 BINS, WITH FIVE PAIRS PER INPUT ON XEON GOLD 6148. M IS SHTPC MASKED-MOMENT BINDING. V IS CONSTRUCTED POSEIDON VECTOR BINDING [21]. RATIOS ARE PAIRED V/M MEDIANS. RSS DENOTES RESIDENT SET SIZE.

Metric	Method	Nonzero	Zero	Near cap
fullProve median (s)	M	3.454	3.447	3.460
	V	310.415	310.741	310.925
Minimum (s)	M	3.444	3.429	3.450
	V	310.084	309.901	309.878
Maximum (s)	M	3.469	3.475	3.524
	V	311.296	311.963	312.015
Paired V/M		**89.91**	**89.90**	**89.83**
RSS median (GiB)	M	0.296	0.298	0.297
	V	13.916	13.870	14.142

while native moment evaluation scans the share vector in $O(D)$.

The constructed vector alternative (Method V) uses blinded Poseidon folds over each role's share and context to bind coordinate-wise share reconstruction to the authorized one-hot (including the sentinel), retaining the hidden-state policy, ticket binding and canonical receiver decoding while reducing public signals from 25 to 22. Evaluated outside live Janus, Table 6 fixes history, coordinates, contribution bits, context and both retained shares across three synthetic inputs covering nonzero contribution, zero at the sentinel, and a legal move 3,299 m from its K-anchor under the 3,300 m cap. Setup verification precedes timing, snarkjs fullProve includes witness generation, and all 30 Xeon proofs, 20 Raspberry Pi 4B proofs and warmups pass independent verification. The complete M and V transition circuits contain 13,327 and 851,715 constraints, respectively. On server hardware, Method M proves transitions in a median 3.45 s within 0.30 GiB RSS, achieving an 89.83–89.91× speedup over Method V (310.4–310.9 s, 13.9–14.1 GiB RSS). On the 1.8 GiB Raspberry Pi 4B (single Cortex-A72 core), Method M proves the three transitions in medians of 11.255 s, 11.238 s and 11.236 s within 0.273–0.278 GiB RSS (and genesis in 4.031 s within 0.185 GiB), whereas Method V requires 13.87–14.14 GiB RSS and exceeds edge memory by over 7.5×. Non-ZK client assembly on the Raspberry Pi 4B takes a median 32.30–32.43 ms (25.71–25.82 ms for Prio3Histogram sharding and 6.54–6.63 ms for HPKE sealing, hashing and Ed25519 signing). Relative to the 167,542 B (163.62 KiB) native DAP report wire, the two 148 B HPKE mask sidecars, 128 B binary Groth16 proof and 64 B signature add 488 B (0.29% overhead), or 1,288 B (0.77%) with the 25 binary public signals (2.04% with the full JSON envelope).

Binding cost. At 10,001 bins, masked-moment binding uses 5,779 standalone constraints, accelerates Xeon proving by 89.83–89.91× over Poseidon vector binding, and runs in 11.24–11.26 s within 0.28 GiB on a 1.8 GiB Raspberry Pi 4B with 0.29–0.77% binary wire overhead.

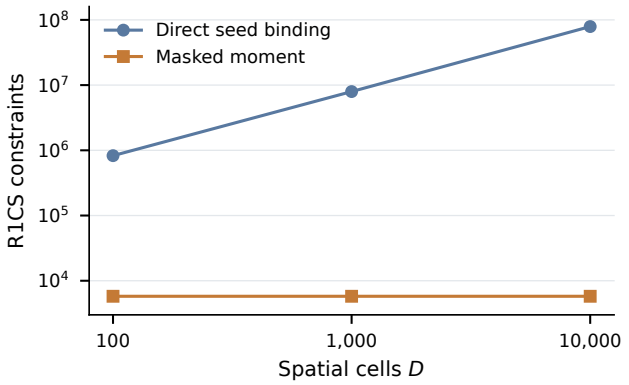


Figure 3. Standalone application binding Rank-1 Constraint System (R1CS) sizes. Complete transitions and native $O(D)$ vector processing are outside this count.

5.4. Feedback and First Exposure

The feedback control demonstrates why source authentication precedes receipt generation. When body-owner signature and authority-certificate checks are disabled, replaying an honest report under an attacker's known mask produces receipts that disclose the private coordinate (revealing cell 4237), an observation that subsequent rejection cannot erase. Enabling the complete guard rejects the forged submission before decryption or feedback, allowing normal reports and exact retries to proceed safely.

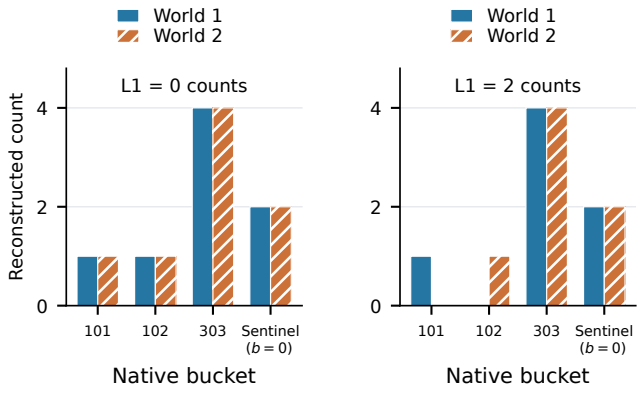
A complete cohort aggregate can conceal differences that become distinguishable in an incomplete subset. Figure 4 illustrates two legal eight-owner histories that yield identical complete heatmaps but diverge by two counts in unnormalized L_1 error across a seven-member subset. The horizontal axis displays the active spatial cell identifiers (cells 101, 102 and 303 in the grid) alongside the sentinel bin ($b = 0$). The native subset control confirms that SHTPC halts export before any honest share is released, enforcing Proposition 1.

External controls exercise alternative aggregation interfaces. The constructed SCIF acceptance bridge detects omitted positives, eligible zeros and equal-count substitutions after trusted reconstruction of an output without external member evidence [11]. A PEAR run reconstructs the exact eight-report, 10,001-bin output and rejects invalid inputs [12], though supplying neither an external accepted-history head nor a locked cohort reference.

Feedback and first exposure. Source authentication blocks receipt-transplant leakage, and pre-export reservation prevents a distinguishing seven-of-eight subset exposure before any honest share becomes decryptable.

5.5. History Recovery

Durable redelivery restores complete cohort outputs after accepted decisions are withheld. In the 100-client campaign covering 700 transitions, withholding one or ten decisions



(a) (b)

Figure 4. Native control outputs for (a) the complete eight-member cohort and (b) the seven-member subset used for separation. The horizontal axis indicates active spatial cell identifiers (101, 102, 303) and the sentinel bin ($b = 0$). The protected SHTPC path rejects the subset before exposure.

TABLE 7. NATIVE CONTROLS EXERCISING THE SECURITY OBLIGATIONS. ROWS DESCRIBE SEPARATE FIXTURES AND SCALES.

Boundary	Fixture / scale	Observed outcome
P1	Cell mismatch control	Cell index 2312/42 mismatch rejected, no commit
P2	Source-feedback control	Guard-off revealed cell 4237, guard-on blocked
P3	Two-owner trace	Conflicting successor rejected, retry idempotent
P4	100 owners, 700 transitions	Missing 1 or 10 blocked, 100-member output restored
P4	Eight owners, seven exposed	Distinguishable seven-of-eight subset blocked

delays release until redelivery restores all 100 eligible members and the exact 10,001-coordinate output. Affected services restart on retained databases while the key-holding controller remains live. Table 7 summarizes the observed security obligations.

A two-owner native trace examines whether retries and selective delivery preserve committed history and the frozen release set. The conflicting successor is rejected, and redelivery restores the frozen set and exact output. This sequential trace complements recovery tests of concurrent preparation, changed peer bytes, arranged commit-first and freeze-first Structured Query Language (SQL) orders, and failure paths that retain accepted state.

A finite model in the Temporal Logic of Actions (TLA+), checked with the TLC model checker (version 2.19), complements these traces across 41 configurations spanning one to three owners and reports, two roles, and one or two windows. The six safety and one fair-progress configurations pass, while guard-removal configurations expose expected invariant violations. Reachability and temporal controls exercise witnesses and fairness assumptions, with model bounds detailed in the artifact.

TABLE 8. SELECTED PHASE AND SESSION COSTS FOR FIVE FULL AND TRUSTED-GENERATION PAIRS.

Phase	Lifecycle time (s)	
	Full (SHTPC)	Trusted generation
History initialization median	3525.968	2504.718
Range	3498.93–3529.40	2492.19–2550.07
Of which setup and genesis	197.245	197.438
Eligible window median	548.266	381.117
Range	544.08–552.05	374.77–381.98
Total session median	**4351.280**	**3156.829**
Range	4323.88–4352.87	3149.91–3205.87

Setup and genesis are included in history initialization. The window includes collection. Total session also includes harness initialization, resource snapshots, post-window observation and harness completion. Unlisted remainder medians are 277.245 s (Full) and 276.600 s (T). Phase medians are computed separately and need not sum to session median.

History recovery. Durable redelivery restores exact 100-member outputs after 1 or 10 withheld decisions. Two-owner Janus traces and 41 bounded TLC configurations check retry idempotence and conflict rejection.

5.6. Cost of the Enforced Contract

Table 8 reports five alternating Full/T pairs on the synthetic 100-client workload. All ten runs satisfy semantic, closure, metering and mode-control checks. Full completes the eligible batch in a median 548.266 s (versus 381.117 s for T) and the total fresh-history session in 4351.280 s (versus 3156.829 s for T), representing paired overheads of 1.44× on the eligible window and 1.38× on the session (paired T/Full medians 0.6958 and 0.7255), where T trusts plaintext report generation and omits transition proofs and moment checks. Within the eligible window for T, the median driver-phase observations are 92.414 s for report construction and registration, 105.581 s for native drive, 41.403 s for native freeze, and 61.555 s for collection. These separately observed phase medians include driver synchronization rather than summing to a single critical path, with handoffs accounting for the remainder.

Comparing Full's 548.266 s window with T's 381.117 s yields a 167.149 s difference. Under ideal 16-way client parallelism, 100 transition proofs at 3.45 s each take about 21.6 s ($100 \times 3.45/16 \approx 21.6$ s), indicating that client proving explains only part of the gap. The remaining difference reflects server-side proof verification, sidecar ingestion, statement registration and receipt serialization across a fresh unamortized history.

Contract cost. Across five 100-client pairs, Full takes a median 548.3 s per eligible batch and 4351.3 s per session (1.44× and 1.38× plaintext trusted generation T) to verify hidden-state transitions and masked moments.

6. Related Work

Three dimensions position SHTPC among related private aggregation systems. (1) *Hidden-state history* verifies sequential per-identity state transitions (e.g., speed and anchor bounds) in zero knowledge. (2) *Width-independent native-output binding* connects client ZK claims to native secret-shared outputs (such as Prio3/VDAF shares) with circuit size independent of D and without in-circuit seed expansion. (3) *First-exposure subset defense* freezes complete eligible cohorts before any honest share becomes decryptable. SCIF [11] enforces multi-party computation (MPC) input inclusion on memoryless inputs without client history policies or width-independent native bridges. PEAR [12] evaluates static nondeterministic polynomial-time (NP) predicates without atomic history heads, width-independent native binding, or first-exposure locking. Ball et al. [14] track global aggregate state rather than private per-client histories, leaving secret-share binding and subset exposure unaddressed. Among the surveyed systems, only SHTPC covers all three dimensions jointly.

6.1. Input validity, inclusion and verifiable aggregation

VDAFs and DAP [4, 5, 22, 23] decouple input validation from aggregation, extending to heavy hitters and streams [24–26]. ELSA, RoFL and fully linear probabilistically checkable proofs (PCPs) enforce input validity on shared data [9, 10, 27], while VerSA and VeriFL verify aggregate correctness [13, 28–30]. These systems validate each input against a static predicate. Commit-and-prove SNARKs such as LegoSNARK [31] link proofs to committed data, while collaborative zk-SNARKs [32] generate proofs over distributed witnesses. Applying these primitives to native aggregation requires interfaces connecting the proven data to the retained shares, state transitions and complete release set. SHTPC bridges ZK state proofs to native secret shares via masked one-hot index moments with width-independent circuit cost under honest service execution.

6.2. Stateful workflows, credentials and membership

Proof-carrying data [33, 34] propagates computation history, while Privacy Pass, DAP report authentication, zk-creds and ZEBRA authorize credentials [35–38]. In location services, VPriv, PrETP, PrivStats and ZKLP enforce charges, upload quotas or regional membership without revealing plaintext trajectories [1–3, 39]. However, static credentials do not prevent replay against stale state, whereas SHTPC atomically authorizes successors against the actual per-identity history head. Unlike federated aggregation schemes that tolerate dropouts via dynamic subsets [40–43], SHTPC enforces strict release, freezing complete eligible cohorts before first share exposure and handling transport faults through durable server redelivery.

6.3. Output privacy

Cryptographic aggregation shields individual payloads and intermediate protocol views from untrusted aggregators, yet the resulting cohort total can still disclose sensitive information across correlated queries. Differential privacy [44, 45] provides orthogonal protection by injecting calibrated perturbation into aggregate histogram releases. Central differential privacy adds noise to the reconstructed heatmap upon decoding, whereas distributed perturbation generates secret-shared randomness across aggregation roles prior to export. Because the quota invariant in Eq. (11) limits each identity to at most one unit contribution per window, the L_1 sensitivity of a released histogram remains bounded, simplifying noise calibration, preventing dropouts from altering the accounted cohort, and complementing protocol-view simulation across release windows.

7. Discussion and Conclusion

This paper incorporates hidden-state validation into native private aggregation. SHTPC binds each admitted state transition to the contribution retained by aggregation roles using native one-hot validity and a compact masked-moment relation, extended to complete-cohort first exposure via atomic admission and durable reservation. End-to-end Janus controls exercise these interfaces, while component benchmarks and lifecycle measurements quantify their costs.

The design targets coordinated sampling applications requiring complete cohort statistics with delayed batch release, preserving strict membership without silent dropout or intersection release while durable server redelivery handles client disconnections. All guarantees apply to admitted claims under the stated integrity and privacy corruption models without establishing physical GPS truth or Sybil resistance, and differencing attacks on exact cohort totals require orthogonal output-level differential privacy.

Future Work. One promising direction extends the architecture to continuous sliding-window streaming and adaptive network recovery while preserving cohort synchronization. Another generalizes the masked-moment binding relation to high-dimensional workloads and applies recursive zero-knowledge proofs to accelerate client proving.

8. Artifact Availability

The repository at https://anonymous.4open.science/r/semantic-artifact-F925/ contains code, proofs and data, with extended bounds, simulations, hybrids and multi-window proofs under artifact/proof/.

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Appendix A.
Interface, Policy and Composition Details

Interface and policy. An independent acceptance rectangle with valid pairs for two primaries also contains their mismatched cross-spliced pair. Thus every rectangle contained in the same-primary relation has one primary. For complete legal and illegal views X_0, X_1, let $a_i = \Pr[V(X_i) = 1]$. Completeness and soundness errors satisfy $e_c + e_s = 1 - a_0 + a_1 \geq 1 - |a_0 - a_1|$, which is one for identical views.

Trace preprocessing and ethical considerations. Both T-Drive and Porto are publicly available, anonymized research benchmark datasets. Their use adheres to dataset licensing terms and involves no human-subject experimentation, active data collection, or deanonymization. For real-trace replay, local equirectangular projection uses radius 6,371,000 m and centers (116.4, 39.9) for T-Drive and (-8.61, 41.15) for Porto. Coordinates are shifted by 5,000 m and floored to integer meters. Points outside [0, 9999] in either coordinate are excluded from legal histories. The earliest observation per 300-second slot is used without gap filling. Pressure histories require eight consecutive occupied slots and, for Porto, one trip. The evaluated policy uses $K = 6$, a 40 m/s speed tier and a 3,300 m anchor cap.

Hidden-state policy. The hidden ADWC relation bounds squared displacement from the accepted predecessor by $\tau_u^2 \Delta t_t^2$, and from the anchor $\mathbf{a}_t = \mathbf{x}_{\max(0,t-K)}$ by $\min\{\tau_u^2 (\Delta T_t^{(K)})^2, c_{\text{policy}}^2\}$, with speed tier τ_u, anchor cap c_{policy} and $\Delta T_t^{(K)} = K \Delta t_{\text{slot}}$. Enrollment proves a hidden genesis. The first $\max(K, 2)$ accepted steps advance history without aggregation eligibility, and the authority checks the proof's commitment references against actual accepted history. Accepted steps, slots, timestamps and revisions are distinct.

First-exposure separation. Swap two owners' positive cells in metadata-compatible legal worlds. Complete aggregates coincide. A batch containing one swapped owner has different aggregates, so the corrupted role and Collector distinguish when the honest share arrives, proving Proposition 1.

Closure event induction. Registration stores full signed candidates without advancing history. Lock fixes P_w, R_w before private processing, and permits reference only actual committed predecessors and anchors. Commit atomically checks head and quota and records the successor, history and signed decision, preserving the legal transitions and real-time order of Definition 1. Apply changes $N_{S,w}(r)$ from zero to one only for eligible commits, with duplicate retries as no-ops. Reserve checks the registration projections in Eq. (12) and all record links, then stores an immutable selector and sum before send. Recovery preserves reservations. Pre-exposure abort retains partial commits and post-send failure retains the exposed snapshot. Induction gives Eq. (6) at every prefix, and native fidelity gives Eq. (14).

Export transaction refinement. The optimized export transaction refines Definition 3. By monotone role application, each eligible record transitions from pending to applied at most once ($N_{S,w}(r) : 0 \to 1$). Atomically validating the projections $\pi_{\text{reg}}(L_{S,w}) = P_w$ and $\pi_{\text{reg}}(M_{S,w}) = R_w$ against the locked registration graph ensures that the reserved share sum $\mathbf{a}_{S,w}$ incorporates all required members and no extraneous reports. Because the resulting snapshot is durably committed before network transmission, idempotent retries do not modify the released cohort or authorize partial subsets.

Appendix B.
Joint Privacy and Concrete Bounds

Trust and lifecycle. Clients hold report-signing keys, admission holds enrollment and authorization keys, and each role holds its input HPKE and receipt keys. The native verification key is uniform in $\{0,1\}^{128}$ and hidden from integrity-game clients. Only the Collector holds the aggregate HPKE secret. The Finalizer uses authenticated admission records, and the Decoder uses the decrypted result and public verification keys.

Authenticated confidential registration binds bodies, permits, commits and receipts to owner, task, window and context. Labeled HPKE protects inputs, sidecars and Collector-bound shares. The Decoder authenticates the finalized-set certificate. Durable journals and leases govern preparation and recovery.

Epochs bind signed context, while identity and quota survive rotation. Revocation, post-compromise recovery and forward secrecy are excluded. The view contains corrupted-role state, excluding honest-role secrets and backups. Durable data survive replay checks and recovery.

BabyJubJub Pedersen commitments [16] use canonical prime-order points, fixed public bases and uniform full-domain blinders. The hidden genesis and transition relations bind the true predecessor and K-anchor. With native field $q = 2^{128} - 7 \cdot 2^{66} + 1$, canonical $r_A, r_R, T \in [0, q-1]$ and $x \in [0, D]$, where $D < q$, give $0 \leq x + r_A + r_R < 3q$. The carry $\chi \in \{0,1,2\}$ therefore makes $x + r_A + r_R = T + \chi q$ an integer equality. Both sides are below $3q < 2^{130} < p_{\text{proof}}$, where the BN254 scalar modulus satisfies $2^{253} < p_{\text{proof}} < 2^{254}$. Thus equality in the proof field cannot wrap. Successful counts fit a signed 64-bit integer, so reconstructed histograms do not wrap modulo q.

Lemma B.1 (Origin isolation)**.** *Outside authentication, digest and origin-allocation failures, a protected ciphertext enters private processing only in its registered source body.*

Proof. Allocation binds the identifier to its owner before publication. The complete body, encryption labels, installed lock and process permit fix the input before decryption. A transplant changes authenticated bytes or requires an already allocated identifier. Exact retries reuse the source and durable native state. □

Source family and extraction budget. Fix finite budgets before the challenge. All counts include honest construction, rejected inputs, offline queries, retries and rolled-back work. The inventory contains four keys over one phase-one setup (current genesis, current transition, historical genesis and the

TABLE 9. ADJACENT HYBRIDS AND THEIR LOSS. N COUNTS HONEST NATIVE SOURCES AND Q_i THE SEVEN STREAM BUDGETS.

Step	Change and simulator knowledge	Loss
1	Nonce deduplication over public unprepared sources	$N^2/2^{128}$
2	Initial usage, seed and binder programming	$N[Q_4 + 2(Q_3 + Q_6 + Q_7) + Q_5]/2^{128}$
3	Share and part programming with corrupt shares fixed	$N[Q_1 + Q_2 + 2Q_7]/2^{128}$
4	Prover-coin removal for unexposed seeds	$NQ_4/2^{128}$
5	Source simulation, full verifier, joint receipts, raw trace and first exposure	0
6	Strict-time truncation after $L + 128$ candidates	t_N

eight-client control (N8)) and two auxiliary $D = 10{,}001$ keys over another independent setup. All six phase-two scalars are nonzero and independent. No target public-wire point divided by its phase-two scalar is exposed. Formal degree bounds across all setup keys ensure that polynomial identities hold over the field. Let $S = S_g + S_t + S_l + S_{\mathrm{ext}}$ count fresh source proofs, and let $V = V_g + V_t + V_l$ count target-key verifications. The $Q_{G_1}, Q_{G_2}, Q_{G_T}, Q_{\mathrm{pair}}$ budgets count typed group-oracle outputs beyond sources, imports and verifications. The counts I_1, I_2 cover source-group imports. The proof-field ledger is

$$\begin{aligned}
h_1 &= 65293168 + 2S + Q_{G_1} + I_1 \\
&\quad + 13V_g + 53V_t + 13V_l, \\
h_2 &= 16131831 + S + Q_{G_2} + I_2 + V, \\
h_T &= 8 + Q_{\mathrm{pair}} + Q_{G_T} + 7V, \\
d_G &= 8388612\tbinom{h_1}{2} + 4194303\tbinom{h_2}{2} + 12582915\tbinom{h_T}{2}, \\
g_{\mathrm{coll}} &= \min\{1, d_G/(p_{\mathrm{proof}} - 4194305)\}, \\
g_{\mathrm{aff}} &= \min\{1, S(3p_{\mathrm{proof}} - 2)/p_{\mathrm{proof}}^2\}, \\
g &= \min\{1, g_{\mathrm{setup}} + g_{\mathrm{coll}} + g_{\mathrm{aff}}\}. \qquad (17)
\end{aligned}$$

Here g_{setup} is the probability that setup sampling yields a zero phase-one evaluation point, α, β or phase-two delta, or an excluded evaluation root. The numerical example includes this term, $g_{\mathrm{setup}} \approx 1.9237 \cdot 10^{-70}$, under independent full-field sampling. For example, with $Q_{G_1} = Q_{G_2} = Q_{G_T} = Q_{\mathrm{pair}} = 2^{32}$, $S_g = S_t = S_{\mathrm{ext}} = V_g = V_t = 2^{19}$, $S_l = V_l = 0$, and $I_1 = I_2 = 2^{20}$, the stated setup and source assumptions give $g \le 2.6738 \cdot 10^{-50} < 2^{-150}$.

Native validity and key-tag assumption. Let Q_i count native stream queries by domain, in order measurement, proof share, joint randomness, prover randomness, query randomness, joint seed and joint part. Put $Q = \sum_{i=1}^{7} Q_i$, and let P_{att} count distinct attack preparation inputs. The actual public key tag is a 96-bit prefix of the verification-key hash. For G_5 external candidate-key queries, let K be the uniform 128-bit key and $Z = f(K)$ its tag. On the no-hit branch, each tag value determines at most G_5 candidate keys E_z. Averaging over tags gives

$$\Pr[K \in E_Z] = 2^{-128} \sum_z |f^{-1}(z) \cap E_z| \le 2^{96} G_5/2^{128}.$$

Thus $\kappa_{\mathrm{tag}} \le \min\{1, G_5/2^{32}\}$, without assuming uniform tag fibers. At $G_5 = 2^{31}$, the term $2\kappa_{\mathrm{tag}}$ already saturates the information-only bound. The factor two charges the two secret-key replacement hybrids. Primitive advantages remain symbolic. An additional SHA256 one-wayness assumption on uniform 128-bit inputs gives the alternative bound $\kappa_{\mathrm{tag}} \le a_{\mathrm{OW128}}$, obtained by scanning all queried keys with one additional hash per candidate. Collision resistance alone does not supply it. The native validity term is

$$\begin{aligned}
v_N = \min\{1, 10256(Q + P_{\mathrm{att}})/q + 2\kappa_{\mathrm{tag}} \\
+ (2P_{\mathrm{att}} + Q)^2/2^{128}\}. \qquad (18)
\end{aligned}$$

Here $10256 = 10002 + 254$. The 10,001-bin invalid-input identity contributes at most $(10001+1)/q$ and a false gadget polynomial of degree at most 254 contributes $254/q$. The root-abort branch remains in the experiment. This invalid-input acceptance term is separate from the nonce-based privacy bound below, since a corrupted role already knows the verification key.

Exposure family.

Definition B.1 (Exposure equivalence)**.** Using the declared indices of Definition 6, legal histories with common public structure μ are exposure-equivalent for R_w when

$$\sum_{i \in I_w} \mathbf{e}_{y_i^{(0)}} = \sum_{i \in I_w} \mathbf{e}_{y_i^{(1)}}. \qquad (19)$$

Authority-only has no aggregate query and does not require this equality.

Proposition B.1 (Effect of the exposure family)**.** *A fixed full cohort permits histogram-preserving reassignment. If every size-k subset can instead be exposed, $1 \le k < n$, equal sums on all such subsets force identical per-record native indices.*

Proof. Set $d_i = \mathbf{e}_{y_i^{(0)}} - \mathbf{e}_{y_i^{(1)}}$. For $k = 1$, each singleton immediately gives $d_i = 0$. For $2 \le k < n$, choose a common $k-1$ element set disjoint from i, j. Exchanging i, j in a size-k subset gives $d_i = d_j$. A size-k sum then gives $kd_i = 0$, hence $d_i = 0$ over integers. □

In the proof hybrid, full-domain blinding ensures language membership for any fixed legal reference history, with history-node aliases reusing the same point. Equation (15) gives the joint receipt law $(T, m_h) = (U, U - F(v_c^{\mathrm{orig}}) - r_c)$. For authority-only, uniform T, U give $(T, m_A, m_R) = (T, U, T - U)$.

The native model is a domain-separated programmable random oracle indexed by usage, seed, complete binders and raw offset. Repeated descriptors also reuse rejected candidates and unconsumed prefetch. Applying these bounds to

the concrete TurboSHAKE128 extendable-output function (XOF) requires this idealization beyond XOF pseudorandomness. The native kernel specializes the two-party, single-proof analysis in the 2024-09-25 revised full version of [23] to the pinned Histogram interface.

Lemma B.2 (Concrete Histogram preparation view)**.** *The two-role, single-proof 10,001-bin Histogram has a 455-element proof, 202-element verifier and 128 interpolation points Ω. For valid inputs and query $t \notin \Omega$, its full verifier is $(0, W_0, \ldots, W_{199}, \sum_{j=0}^{99} W_{2j} W_{2j+1})$, with independent uniform W_i. Queries in Ω take the same root-abort branch in both worlds.*

Proof. Each valid-input wire has nonzero random-value coefficient $(t^{128} - 1)/(128(t - 1))$. Its affine inverse makes evaluations independent uniform W_i, determining the gadget coordinate. For $t \in \Omega$ both worlds abort identically. □

Theorem B.1 (Concrete two-party native privacy)**.** *In this programmable-oracle model, let N count honest native source reports, including unprepared reports. A single-world native stream simulation has loss at most*

$$b_N = \min\{1, (N^2 + NQ_*)/2^{128}\}, \tag{20}$$

where $Q_* = Q_1 + Q_2 + Q_5 + 2(Q_3 + Q_4 + Q_6) + 4Q_7$. Under Definition 6's exposure-time interface, the simulator queries one complete cohort when the first honest share becomes decryptable and none earlier, under Eq. (19)*.*

Proof intuition. Fix either corrupt role's original measurement/proof shares, key and nonce. Program fresh domain-separated streams, charging collisions and prior queries to Eq. (20). Sampling Σ from the verifier distribution and setting $z_h = \Sigma - z_c^{\text{orig}}$ makes the simulated honest share statistically identical to real execution. Active responses z_c' are verified via $\mathsf{Decide}(z_h + z_c')$ and seed checks. Helper retries replay journaled bytes, while Leader recomputations preserve Σ, z_h and commit at most once, ensuring consistent simulation across retries without online verifier oracles. □

Theorem B.2 (Joint privacy of admission and native release)**.** *Use Definition 6's privacy experiment, source and oracle models, and primitive bounds. Let X denote authority-only, active-role-and-Collector, or that aggregation view with authority business records. Let g be Eq.* (17)*'s bound, ϵ_H the full SHA256 collision advantage, a_X correct-party signature existential unforgeability under chosen-message attack (EUF-CMA) advantages, and c_B Pedersen binding advantage. Let N_{id} count honest identifier draws and G_{id} adversarial guesses. With C_X hidden HPKE encapsulations and labeled indistinguishability under chosen-ciphertext attack (IND-CCA) advantages H_X, set*

$$B_X = \min\{1, g + \epsilon_H + a_X + c_B + N_{\text{id}} G_{\text{id}}/2^{128}\}, \tag{21}$$

$$e_X = \min\{1, 2(C_X + 10)B_X + H_X + b_N + t_N\}, \tag{22}$$

$$\text{Adv}_X \le \min\{1, 2e_X\}, \tag{23}$$

where $t_N \le \min\{1, 5N\binom{10129}{129}\rho^{129}\}$ and $\rho = (2^{128} - q)/2^{128}$. Authority-only needs no aggregate equality, whereas the other two views require Eq. (19)*.*

Hybrid outline. Rows 1–4 of Table 9 sum to the bound in Eq. (20). The C_X labeled HPKE hops and ten macro hops each have two endpoints, each charged B_X. Their endpoint charges give $2(C_X + 10)B_X$, while H_X adds one IND-CCA advantage per hidden encapsulation, replacing honest ciphertexts and simulated receipts step by step. The five field streams provide measurement, proof share, joint, prover and query randomness, while separate seed and part streams generate pseudorandom bytes. Idempotent retries preserve report identifiers, cryptographic keys, nonces, shares and sampling randomness across changing network responses. Repeated stream descriptors reuse their initialized generator state. Truncating after $L + 128$ candidates requires 129 consecutive rejections to induce failure, where the union bound over $5N$ streams with $L \le 10001$ yields bound t_N. Because the two challenge histories share one ideal view, their distinguishing distances add. □

Aggregation views have $C_X \le 2N$, with one honest-role input and sidecar per honest source. Authority-only has $C_X \le 4N + E_{\text{enc}}$ for both roles and fresh visible aggregate encapsulations, including re-encrypted retries. Each H_X term uses one recipient-key-specific challenge with finite time, decryption-query and byte budgets, where adversary running time accounts for reduction execution, and the integrity reduction route relies on bound B_C.

Theorem B.3 (Predeclared multi-window privacy)**.** *Use Definition 6 with the predeclared k-window contract of Section 2.1 in one task, profile and key namespace. For P and $P{+}A$, require Eq.* (19) *for every declared complete cohort, including reserved unexposed cohorts. A joint simulator retains shared history, source, feedback and retry state. Persistent reservations fix each cohort and payload. Unlike Definition 6, aggregation-view simulators can privately prefetch after reservation, before the first honest share becomes decryptable, and cache through recovery. Reports remain single-window, while accepted-state references can cross windows.*

The authority-only analysis retains the same bound form, with resource budgets evaluated over the joint multi-window (MW) execution. For each observer,

$$e_X^{\text{MW}} = \min\{1, 2(C_X^{\text{MW}} + 10)B_X^{\text{MW}} + H_X^{\text{MW}} + b_N^{\text{MW}} + t_N^{\text{MW}}\},$$

and $\text{Adv}_X \le \min\{1, 2e_X^{\text{MW}}\}$. With joint native source count N_{MW}, $C_P^{\text{MW}} = C_{P+A}^{\text{MW}} = 2N_{\text{MW}}$ and $C_A^{\text{MW}} = 4N_{\text{MW}} + E_{\text{enc,A}}$, counting all fresh authority-visible aggregate encapsulations in $E_{\text{enc,A}}$. The deployed contract remains single-release per task and key.